



Scotch-Weld™ One-part Epoxy Adhesive

6011 low viscosity White, 6011 White, 6011NS Black, 6011NSTF White

Technical Data

Dec. 2015

Product Description	Scotch-Weld™6011 low viscosity White, 6011 White, 6011NS Black and 6011NSTF White is high performance one-part low temperature fast cure epoxy adhesive with excellent shear strength and good heat and humidity resistance. It provides good adhesion strength to various types of substrates when cured and is designed for heat sensitive electronics and semiconductor devices, such as memory cards, CCD/CMOS, LED lens, earphone and high throughput display assembly etc.			
Features	• One part • Cure at low temperature • Fast cure	• Excellent shear strength • Excellent heat and humidity resistance • Excellent sealing • Good Pot life		

Typical Uncured Physical Properties

Note: The following technical information and data should be considered representative or typical only and should not be used for specification purposes.

Note: **It may be take different time to reach cure temperature in different process. Sufficient time must be added to allow the bond location to reach the desired cure temperature. Curing condition should be developed for each device.**

	6011 low viscosity White	6011 White	6011NS Black	6011NSTF White
Halogen content	<900 ppm	<900 ppm	<900 ppm	<900 ppm
Viscosity (Brookfield, #4 Spindle, 20 rpm at 23°C)	4500cP (50RPM)	18000cP	32000cP	40000cP
Thixotropic	No, self-leveling	No	Yes	Yes
Density g/cc	1.10 - 1.30	1.30 – 1.60	1.10 – 1.50	1.10 – 1.50
Color	White	White	Black	White
Cure profile	60°C	15 mins	15 mins	20 mins
	90°C	2 mins 30 secs	2 mins 30 secs	3 min
Pot life at 23°C	24 hours	24 hours	24 hours	24 hours
Shelf life at -20°C	12 months	12 months	12 months	12 months

Pot life is defined as viscosity up <=50% after store at 23°C.

Typical Cured Physical Properties

Overlap shear test (ASTM D 1002-72)

Overlap shear (OLS) strengths were measured on 1 in. wide 1/2 in. overlap specimens. The thickness of the bondline was 0.005-0.008 in. All strengths were measured at 73°F (23°C) except where noted. The separation rate of the testing jaws was 0.1 in. per minute.

	6011 low viscosity White	6011 White	6011NS Black	6011NSTF White
Shore D Hardness	78	78	78	75
Overlap shear strength Abraded Al to Abraded Al	18 MPa	18 MPa	18 MPa	18 MPa
Overlap shear strength Abraded Al to PC/ABS	13 MPa	13 MPa	13 MPa	13 MPa
Aging 10 days at 65°C, 95%RH Abraded Al to PC/ABS	8.5 MPa	8.5 MPa	8.5 MPa	8.5 MPa

Handling/Curing

Directions for Use



Scotch-Weld™ One-part Epoxy Adhesive

6011 low viscosity White, 6011 White, 6011NS Black,
6011NSTF White

Information

Handling/Curing Information (continued)

1. Storage at -20°C
 2. Warm products to room temperature before opening containers to restore proper application consistency and to prevent moisture condensation on adhesive surface. Containers may be stored at room temperature for 1-2 hours to thaw. Do not warm at temperatures above 27°C.
 3. Use glove to avoid contacting container directly.
 4. Avoid freezing products again after warming
 5. Recommended cure profile: 30mins at 60°C, 20mins at 70°C, 15mins at 80°C, 10mins at 90°C.
- Curing condition should be developed for each device.**

Surface Preparation

The following cleaning methods are suggested for common surfaces:

Steel:

1. Wipe free of dust with oil-free solvent such as Methyl Ethyl Ketone.*
2. Sandblast or abrade using clean fine grit abrasives.
3. Wipe again with solvent to remove loose particles.

Aluminum:

1. Vapor Degrease – Perchloroethylene* condensing vapors for 5-10 minutes.
2. Alkaline Degrease – Oakite 164 solution (9-11 oz./gallon water) at 190°F ± 10°F (87°C ± 5°C) for 10-20 minutes. Rinse immediately in large quantities of cold running water.
3. Acid (FPL) Etch – Place panels in their following solution for 10 minutes at 150°F ± 5°F (66°C ± 2°C).
Sodium Dichromate 4.1-4.9 oz./gallon
Sulfuric Acid, 66°Be 38.5-41.5 oz./gallon
2024-T3 aluminum (dissolved) 0.2 oz./gallon minimum
Tap water as needed to balance
4. Rinse – Rinse panels in clear running tap water.
5. Dry – Air dry 15 minutes; force dry 10 minutes at 150°F ± 10°F (66°C ± 5°C).
6. If primer is to be used, it should be applied within 4 hours after surface preparation.

Plastics:

1. Solvent wipe with Isopropyl Alcohol.*
2. Abrade using clean fine grit abrasives.
3. Solvent wipe with Isopropyl Alcohol.*

Rubbers:

1. Solvent wipe with Methyl Ethyl Ketone.*
2. Abrade using clean fine grit abrasives.
3. Solvent wipe with Methyl Ethyl Ketone.*

Glass:

1. Solvent wipe with acetone or Methyl Ethyl Ketone.*

Note: For glass applications which will be subjected to high moisture/humidity conditions, 3M™ Scotch-Weld™ Primer EC-3901 should be used to prime the glass.

*Note: Prior to use of these solvents, extinguish or eliminate any ignition sources and read and follow supplier's environmental, health, and safety



Scotch-Weld™ One-part Epoxy Adhesive

6011 low viscosity White, 6011 White, 6011NS Black,
6011NSTF White

	recommendations listed on the MSDS and product label.
Storage and Shelf Life	Storage: Store products at -20° C or below for maximum storage life. Higher temperatures reduce normal storage life. Rotate stock on a “first-in-first-out” basis. Shelf Life: This product has a shelf life of 6 months when stored in their unopened containers at -20°C.
Precautionary Information	Refer to Product Label and Material Safety Data Sheet for health and safety information before using this product. For additional health and safety information, contact 3M
Technical Information	The technical information, recommendations and other statements contained in this document are based upon tests or experience that 3M believes are reliable, but the accuracy or completeness of such information is not guaranteed.
Warranty, Limited Remedy, and Disclaimer	If the 3M product is proved to be defective, The exclusive remedy, at 3M'S option, shall be to refund the purchase price of or to repair or replace the defective 3M product. 3M shall not otherwise be liable for loss or damages, whether direct, indirect, special, incidental, or consequential, regardless of the legal theory asserted, including, but not limited to, contract, negligence, warranty, or strict liability.



3M China Ltd.

Industrial Adhesive and Tape Division

Tel: 021—62753535

Fax: 021—62752343

Website: www.3M.com/cn